OPERATING SYSTEMS – ASSIGNMENT 1

Name: S. Sharvesh **Register No:** 192525458

TASK 4 – Evaluate the Services an Operating System Must Provide (4 Marks)

Evaluate user interface, program execution, I/O operations, file systems, communication, error detection, resource allocation, accounting, protection and security.

Discuss which services can be removed from bedside monitors, which are essential for the database server and how the OS protects one patient's records from another user's programs.

Alternative Concept and Complexity Analysis: If tablets used a full desktop OS instead of a reduced handheld OS, evaluate extra memory/storage, start-up time and battery life.

Evidence Expected: A service table for each environment with justified inclusions and exclusions.

ANSWER

4.1 Services that can be removed or reduced on monitors

Remove general-purpose desktop GUI, multimedia, printing, consumer app store, unnecessary network protocols and non-essential background services.

Keep deterministic scheduling, sensor drivers, alarm output, watchdog, essential local logging, secure boot, authenticated configuration and a minimal secure update mechanism.

Do not remove fault detection, access control or event logging merely to save memory; these support patient safety and traceability.

4.2 Essential database services and record isolation

The database server requires process execution, memory and CPU scheduling, high-performance I/O, reliable filesystems, network communication, error detection, resource accounting, authentication, authorisation, encryption and audit logging. Patient records should be protected through unique user identities, role-based access control, least privilege, database views/row-level security where appropriate, parameterised queries, encrypted connections and encrypted storage. Each application should use a restricted service account rather than a shared administrator account. The OS enforces process isolation using virtual address spaces and file permissions; the DBMS enforces patient-level and role-level permissions.

4.3 Alternative: full desktop OS on tablets

A full desktop OS can provide broad compatibility and familiar management tools, but it adds background services, drivers, larger storage requirements and more memory use. Extra service memory is the sum of the memory used by those services, O(s) for s enabled services in a simple accounting model. Startup time increases as more services initialise and battery life decreases because background CPU, storage and wireless activity consume energy. A managed mobile OS is therefore preferable for 120 tablets used for an entire shift. A desktop OS could be justified only if a required clinical application cannot run on the mobile platform and the tablet hardware has sufficient capacity.

QUESTION FROM THE PROVIDED ASSIGNMENT

OS service | Monitor | Tablet / workstation | Database / imaging server

User interface | Minimal local status and alarm interface; no unnecessary desktop shell. | Touch UI on tablet; full GUI on workstation. | Limited administrator console; server UI not required for normal operation.

Program execution | Only validated acquisition, threshold and alarm tasks. | Clinical apps and managed background services. | Database engine, backup, analytics and image-processing processes.

I/O operations | Sensor and alarm device I/O; deterministic drivers. | Touch, display, Wi-Fi, camera/scanner as authorised. | High-speed disk, network and imaging device I/O.

File system | Small, robust configuration/log storage; avoid unnecessary files. | Encrypted local cache with controlled retention. | Journaled filesystem, database storage, snapshots and backups.

Communication | Only required telemetry; alarm remains local. | Secure hospital Wi-Fi and API access. | Replication, client connections and partner/cloud APIs.

Error detection | Watchdog, sensor plausibility checks, fault and alarm logging. | Application crash reporting, connectivity and battery monitoring. | Disk/RAID/replication monitoring, checksums and service health checks.

Resource allocation | Fixed priorities and bounded CPU/memory allocation. | Power-aware CPU, memory and network scheduling. | CPU, RAM, storage and network allocation; workload controls.

Accounting | Minimal event and alarm audit logs. | User activity and device compliance logs. | Detailed database audit, resource and transaction logs.

Protection/security | Secure boot, least privilege, signed updates, authenticated access. | Encryption, screen lock, MDM, app sandbox and MFA. | RBAC, MFA, encryption, network segmentation, audit and backup protection.